More AI Reviewers Don't Help — Decorrelation Does

A pre-registered, compute-matched comparison of four AI code-review architectures across **2,415 reviews** — including 30 real pull requests from the platform we built for our client.

Companion to: "Evaluating Decorrelation in LLM Code Review: A Pre-Registered Study"

En-Ping Su · Tong Wu · Shiting Huang · Mengshan Li

Khoury College of Computer Sciences, Northeastern University · Vancouver · Advisor: Yvonne Coady · Industry Partner: Indigenomics Institute (Carol Anne Hilton, Founder)

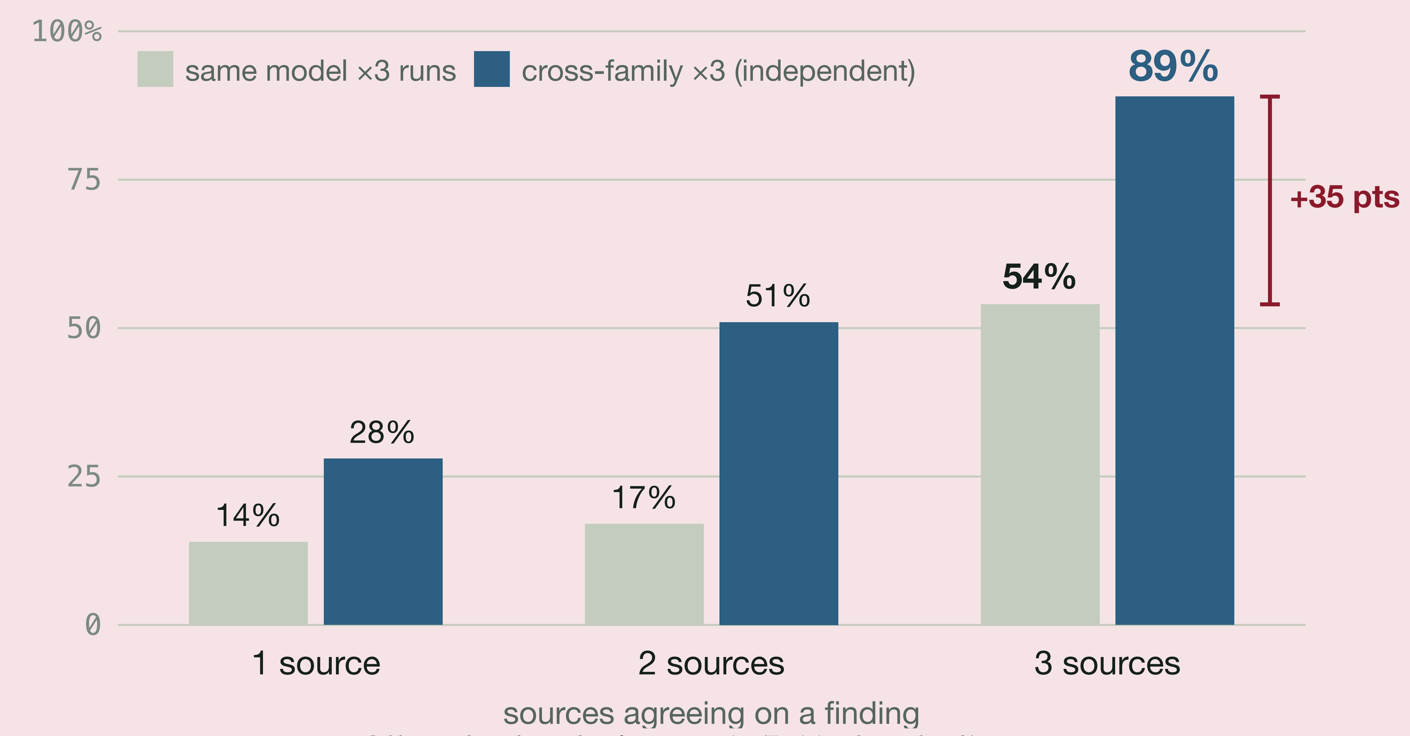
THE CONTRIBUTION

Agreement predicts truth — but only when the reviewers are independent.

Error decorrelation — not more agents, specialists, or debate — is what turns agreement into a real signal.

89% three **independent** model families that agree match injected ground truth

54% one model run three times — agreement is mostly noise



Golden-match rate by number of sources agreeing (E1, injected ground truth).

01 PROBLEM & MOTIVATION

THE CLIENT

The **Indigenomics Institute** needs corporate Reconciliation Action Plan commitments made legible and trackable — and AI now writes much of the software that does it.

THE RESEARCH

Multi-agent AI reviewers are proliferating — managers, specialists, voters — and priced accordingly. Does that **structure** improve review, or just cost more?

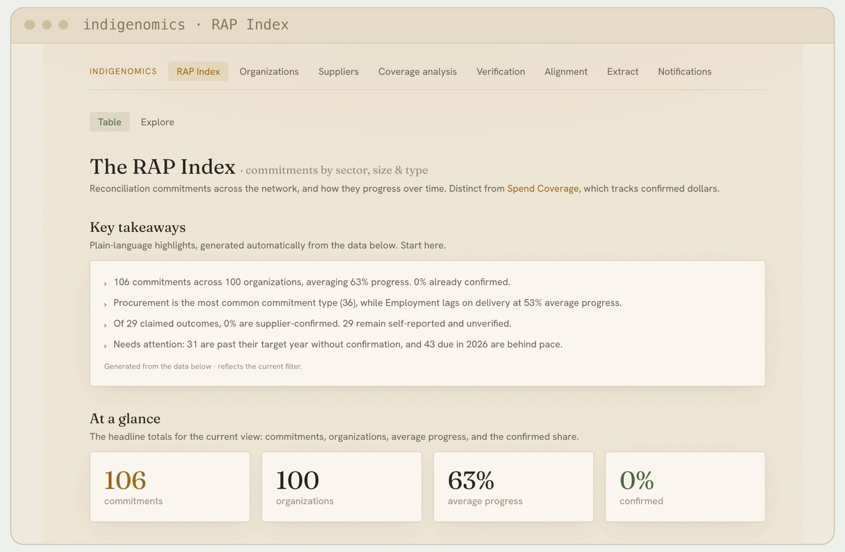
> GROUNDED IN
> Carol Anne Hilton's economic-reconciliation framework — an economic lens, not a case summarizer.

2,415	179	4	×3
reviews	pull requests	architectures	runs each

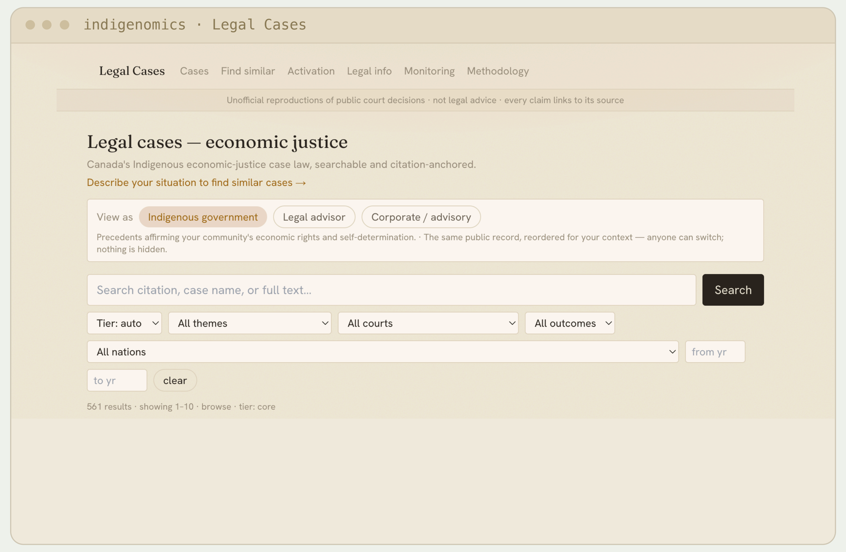
02 APPROACH

WHAT WE BUILT FOR THE CLIENT

The **RAP Data Portal** — one platform, two lenses: corporate reconciliation commitments and Indigenous legal precedents, as comparable searchable data on AWS.



Lens 1 — RAP Index. *106 commitments across 100 organizations, 63% average progress.*



Lens 2 — Legal Cases. *561 Indigenous economic-justice decisions, citation-anchored.*

THE EXPERIMENT — APPARATUS, NOT THE FINDING

Agentless → **Generalists-3** → **Hierarchical** → **Consensus**

Each rung adds exactly one capability — sampling, then specialization, then communication.

Held constant: same PR snapshot · same model (Claude Haiku 4.5) · same prompts · same evaluation pipeline · three runs per arm · **pre-registered on OSF**

E1 · Qodo 99 PRs · ground truth

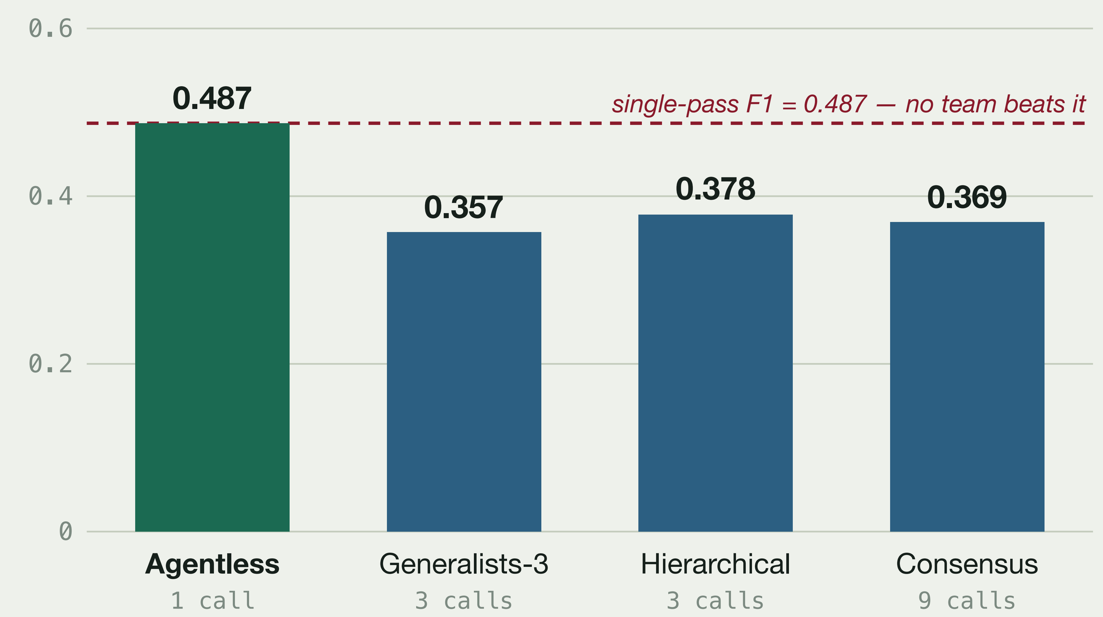
E2 · SWE-PRBench 50 PRs · human review

E3 · our portal 30 PRs · no oracle

03 KEY RESULTS

Structure is null. No multi-agent arm beats single-pass F1; specialists tie compute-matched sampling (p=0.73); peer debate adds nothing over a manager (p=0.12).

Decorrelation pays — on precision, not coverage (89% vs 54%). **Bounded:** recall plateaus at ≈0.83; on the portal's own unlabeled code, LLM judges disagree (κ=0.03).



Semantic F1 by architecture — extra agents raise recall but halve precision, so no team beats single-pass.

04 WHAT THE EXTRA AGENTS ACTUALLY BUY

Adding agents doesn't buy quality — it buys **recall at a price**. Every confirmed true finding costs **0.34** LLM calls under single-pass review and **1.76** under peer consensus, for a **lower-F1** result.

That is the practical case for decorrelation: **spend the budget on independence, not on more of the same reviewer.**

IMPACT — FOR THE FIELD

Don't deploy multi-agent review expecting a quality lift. **Decorrelate your verifiers** — independent model families — and read their agreement as a precision signal.

IMPACT — FOR THE CLIENT

A verifiable, trackable view of economic reconciliation — **commitments + legal precedents in one platform**, delivered to the Indigenomics Institute.

NEXT STEPS

Tool-grounded verifiers — execution and static analysis as a correctness oracle. **Open science:** all runs, ablations & artifacts released for replay.

Acknowledgements. Lino Coria Mendoza (weekly progress check-ins) · Indigenomics Institute: Shawn Anderson (technical contact), Eve Marenghi (scheduling), K. Krasnow Waterman (legal advice) · Compute: AWS Bedrock.

Pre-registered on OSF · github.com/SharonHuang77/rap-review-research